A MULTI-VIEW RETRIEVAL SYSTEM

MULTI-FACETED MUSIC RETRIEVAL.

Combining audio, text, and graph embeddings into a single recommendation system that outperforms any single view.

Team

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Dataset

Free Music Archive · 2,000 tracks

/ 01 — THE PROBLEM

A single embedding can't answer every kind of music question.

QUERY · MOOD

"Chill lo-fi vibes"

requires acoustic features — mood & instrument feel

QUERY · SEMANTIC

"Folk artists about travel"

requires text & lyrics — semantic meaning

QUERY · STRUCTURAL

"Songs like this one"

requires graph relationships — co-genre, co-artist

No single embedding captures all three.

→ build independent views, then fuse

/ 03 — ARCHITECTURE

From raw audio & metadata to fused recommendations.

/ INPUTS

- MP3 audio · 30s clips
- tracks.csv · genres.csv
- Genius API · lyrics

/ EMBEDDING MODELS

- CLAP · HTSAT-tiny — 512-d
- SBERT · all-MiniLM-L6-v2 — 384-d
- OpenL3 · self-supervised — 512-d

256-d

/ FUSION & OUTPUT

- Score fusion · weighted

L2-normalised vectors · cosine via dot product

/ 04 — THE THREE VIEWS

Each view captures a different facet of music.

VIEW 01 · VIBE

CLAP

HTSAT-tiny 512-d · audio contrastive (audio↔text)

High-level semantics — mood, genre feel, instruments.

VIEW 02 · SEMANTIC

SBERT

all-MiniLM-L6-v2 384-d · text siamese transformer

Textual semantics — artist identity, lyrical content.

VIEW 03 · ACOUSTIC

OpenL3

music-pretrained 512-d · audio self-supervised

Low-level acoustics — timbre, rhythm, texture.

/ 05 — DATASET

Free Music Archive — open-licensed benchmark.

2,000
TRACKS · FMA SMALL

8
TOP-LEVEL GENRES

294
ECHO NEST OVERLAP

/ GENRES

Electronic · Experimental · Folk · Hip-Hop · Instrumental · International · Pop · Rock

/ SOURCE

github.com/mdeff/fma · Defferrard et al., 2017

/ ADDITIONAL DATA SOURCES

FMA	tracks.csv, genres.csv — artist, title, tags, genre labels
GENIUS	Song lyrics — first 1,000 characters per track
ECHO	8 continuous audio features — danceability, energy, valence…
EVAL	294-track Echo Nest overlap → independent ground truth

/ SECTION 01

CLAP.

Vibe & text-to-music search — a 512-d shared space for audio and text via contrastive pretraining.

/ 01 — CLAP RESULTS

Natural-language queries against 7,997 audio embeddings.

QUERY	TOP RESULT	GENRE	COS
"sad piano ballad"	DUITA — XPURM	Instrumental	0.65
"aggressive heavy metal"	Dead Elements — Angstbreaker	Rock	0.54
"upbeat happy pop song"	One Way Love — Ready for Men	Pop	0.52
"acoustic guitar folk"	Wainiha Valley — Mia Doi Todd	Folk	0.49

/ GENRE STRUCTURE IN CLAP SPACE

SIM Hip-Hop ≈ Pop · 0.81
Folk ≈ International · 0.80

DIFF Rock ↔ Electronic · 0.64

PCA 50 components capture ~85% of variance

CLAP retrieves on-genre top results across all natural-language queries — even with no fine-tuning.

/ SECTION 02

SBERT.

Lyrics & semantic search — a siamese transformer mapping metadata + lyrics to 384-d embeddings.

/ 02 — SBERT ANALYSIS

A more uniform 384-d space than CLAP's.

/ MODEL

all-MiniLM-L6-v2 — a 6-layer transformer with 384-d output, trained on 1B+ sentence pairs (NLI + paraphrase).

/ PCA VARIANCE

4.1%
TOP COMPONENT

181
COMPONENTS FOR 90%

vs. 50 components for CLAP's 512-d space — SBERT uses its dimensions more uniformly, which is desirable for retrieval.

/ REPRESENTATION TESTS

SEM **Semantic robustness** — only 10% overlap between "lonely" vs "isolated" queries

LEX **Lexical bias** — "Blue music" retrieves titles with "Blue", not the Blues genre

TRC **Truncation** — minimal impact; FMA strings < 50 words

/ METHODOLOGICAL CORRECTION

A leakage path from genre labels into SBERT inputs — caught and fixed.

PROBLEM

Genre leaked through input strings.

"{artist} - {title}. Tags: {genre}, {tags}"

48.8% of tags contained genre-like labels (folk, pop, melodic…). When evaluation uses genre as ground truth, the model matches on the label itself — inflating retrieval scores.

FIX

Three-step cleanup.

01 Removed genre_top from input entirely

02 Filtered all 8 genre words from tags

03 Added Genius API lyrics (1,000 chars) as semantic signal

"{title} by {artist}. Tags: {filtered}. Lyrics: {lyrics}"

→ Genre retained for evaluation only. Never seen by the embedding model.

/ SECTION 03

OPENL3.

Acoustic similarity — self-supervised audio embeddings capturing timbre, rhythm, and texture.

/ 03 — OPENL3

A purely acoustic space — no semantic supervision.

/ MODEL

OpenL3 (content_type='music' , 512-d). Trained via self-supervised audio-visual correspondence on AudioSet — captures low-level acoustic properties without any label supervision.

/ KEY DIFFERENCE FROM CLAP

CLAP	maps audio into a space shared with text — semantic
L3	maps audio into a purely acoustic space — sounds-alike
→	Two songs that sound similar are close — even across genres

USE CASE

"Find songs that sound like this one."

Index all 8,000 tracks via FAISS for instant nearest-neighbour search.

8,000	512	<1ms
TRACKS INDEXED	DIMENSIONS	PER QUERY

/ 06 — WHY FUSION

SBERT and OpenL3 retrieve almost entirely different tracks for the same query.

5.6%

OVERLAP @ TOP-20

Only 1.1 of 20 neighbours shared.

-0.77

SPEARMAN RANK CORRELATION

SBERT-similar tracks are OpenL3-dissimilar.

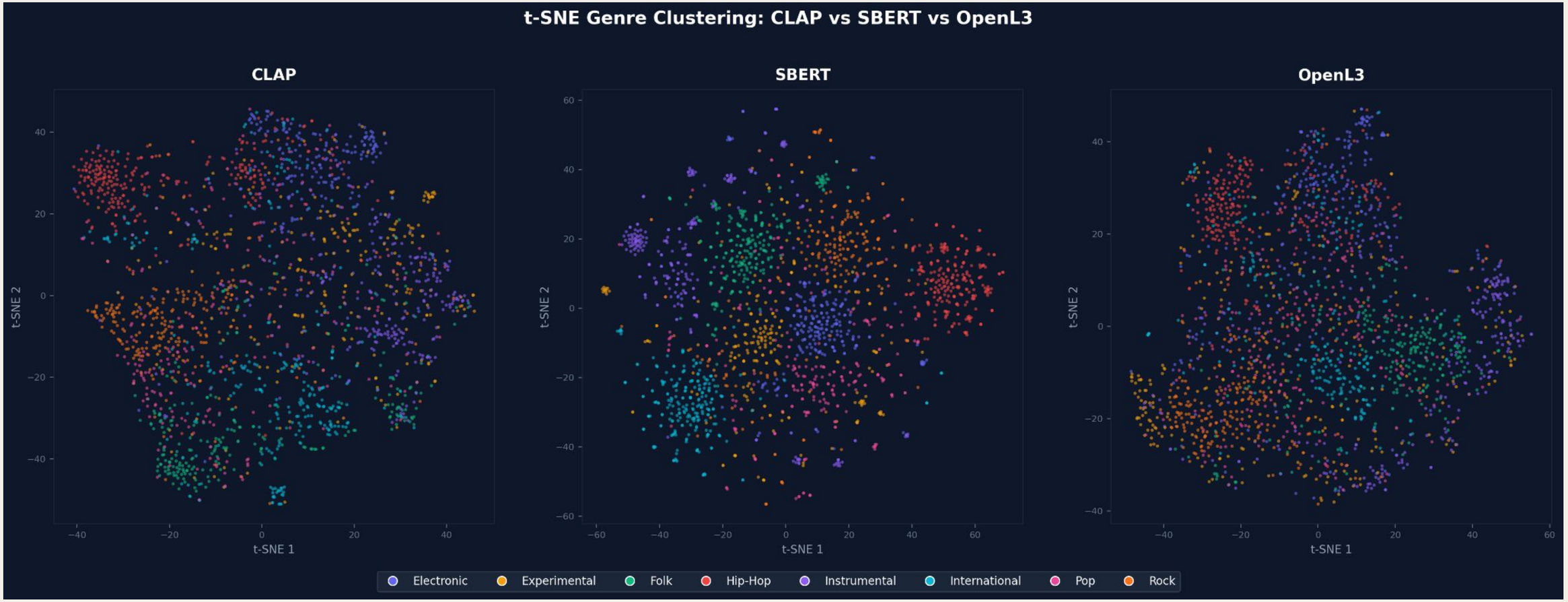
/ INTERPRETATION

Strong complementarity — each modality contributes information the other lacks.

This is the core motivation for multi-view fusion: combining independent signals produces better retrieval than any single view alone.

/ 06 — WHY FUSION

SBERT and OpenL3 retrieve almost entirely different tracks for the same query.



/ 08 — EVALUATION

Independent ground truth — 294 tracks the models never saw.

METHOD	AVG DIST.	VS. RANDOM	P-VALUE
Random baseline	3.80	—	—
SBERT · text + lyrics	3.33	−12.4%	1.6e−06
Fused · SBERT + OpenL3	3.22	−15.1%	1.4e−08
OpenL3 · audio	**2.87**	**−24.3%**	**1.8e−21**

/ KEY TAKEAWAYS

01 OpenL3 wins — both it & Echo Nest live in the acoustic domain

02 Fused beats either modality alone — confirms complementary signals

03 Even SBERT (text only) significantly beats random ($p < 10^{-6}$)

04 All differences statistically significant (paired t-test)

/ 09 — FINE-TUNING

Pull genres apart with FMA-specific contrastive learning.

/ APPROACH

01 Construct (audio, text) pairs from FMA metadata

02 Fine-tune with InfoNCE loss — pull pairs together, push others apart

03 Batch size 32–64 (more negatives = better contrast)

04 5–15 epochs, early stopping on validation loss

/ GOAL — REDUCE GENRE CONFLATION

HIP-HOP ↔ POP
currently nearly identical in CLAP space

0.81

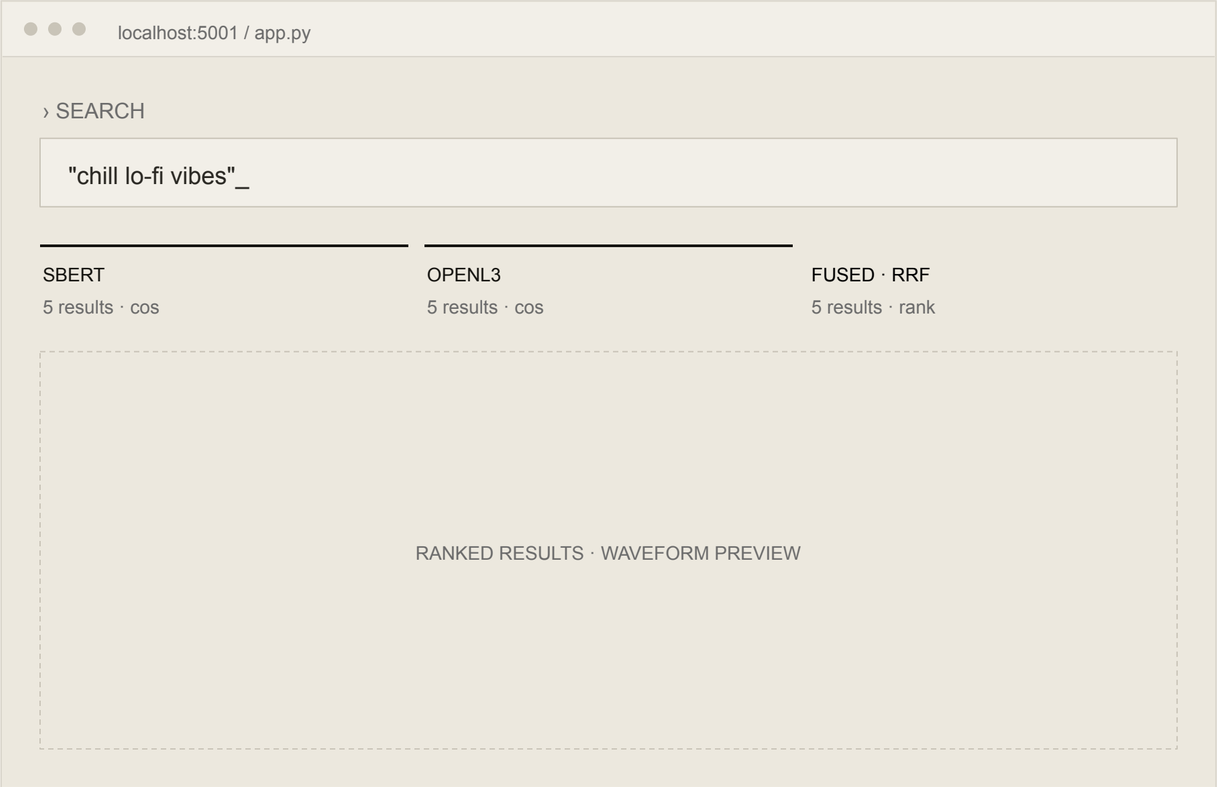
/ BEFORE / AFTER COMPARISON

VIZ t-SNE projections, genre centroid heatmaps

IR P@10, NDCG@10 retrieval quality

/ 10 — DEMO

Multi-view recommendation app — side-by-side, per-view.



/ FEATURES

- → Search any track by title, artist, or genre
- → Per-view recommendations (SBERT · OpenL3 · CLAP)
- → Fused results via Reciprocal Rank Fusion
- → Per-view cosine scores shown alongside
- → Audio playback of original MP3 files

/ TECHNICAL

All embeddings L2-normalised; cosine via dot product. FAISS IndexFlatIP — exact search, <1ms / query @ 8k vectors.

/ 11 — FINDINGS

Four takeaways from the project.

01

Multi-view fusion outperforms any single view.

Fused: −15.1% vs random · text alone −12.4% · audio alone −24.3%.

02

Text and audio are highly complementary.

Only 5.6% top-20 overlap; Spearman ρ = −0.77.

03

Data leakage is a real risk in retrieval evaluation.

48.8% of tags contained genre labels — caught and removed before evaluation.

04

Domain ground truth requires care.

Echo Nest features provide independent, audio-derived ground truth no model saw during training.

/ APPENDIX A — GLOSSARY

Reference terms used throughout.

EMBEDDING
A fixed-size numerical vector representing a track in a continuous space where similar items are close.

COSINE SIMILARITY
Cosine of the angle between two vectors, [-1, 1]. Equals dot product for L2-normalised vectors.

CLAP
Contrastive Language-Audio Pretraining — shared audio/text space; mood, genre, instruments.

OPENL3
Self-supervised audio embeddings on AudioSet — timbre, rhythm, texture (no labels).

RRF
Reciprocal Rank Fusion — sum 1/(k+rank) across views. Robust, no weight tuning.

T-SNE
t-distributed Stochastic Neighbor Embedding — 2D visualisation of high-dimensional vectors.

FAISS
Facebook AI Similarity Search — efficient nearest-neighbour search. We use IndexFlatIP (exact inner product).

CONTRASTIVE LEARNING
Learns to map similar pairs close, dissimilar pairs far. Typically uses InfoNCE loss.

SBERT
Sentence-BERT — siamese transformer for sentence embeddings. all-MiniLM-L6-v2.

GNN
Graph Neural Network — node embeddings from neighbour aggregation. SAGEConv, heterogeneous graph.

DATA LEAKAGE
Eval targets leaking into model inputs, inflating metrics. We removed genre from SBERT input.

INFONCE
Contrastive loss using softmax over (positive vs many negatives). Used for CLAP fine-tuning.

/ APPENDIX B — REFERENCES

Datasets, models & libraries.

/ DATASETS

FMA **Free Music Archive (Small)** 8,000 tracks · 8 genres · ~30s clips · open-licensed. Defferrard et al., 2017 · github.com/mdeff/fma

ECHO **Echo Nest / Spotify audio features** Danceability · energy · valence · tempo · acousticness · … 294-track FMA overlap → independent ground truth

GEN **Genius API** Song lyrics (first 1,000 chars per track).

AS **AudioSet** Training data for OpenL3 (audio-visual correspondence). Gemmeke et al., 2017 · research.google.com/audioset

/ MODELS & LIBRARIES

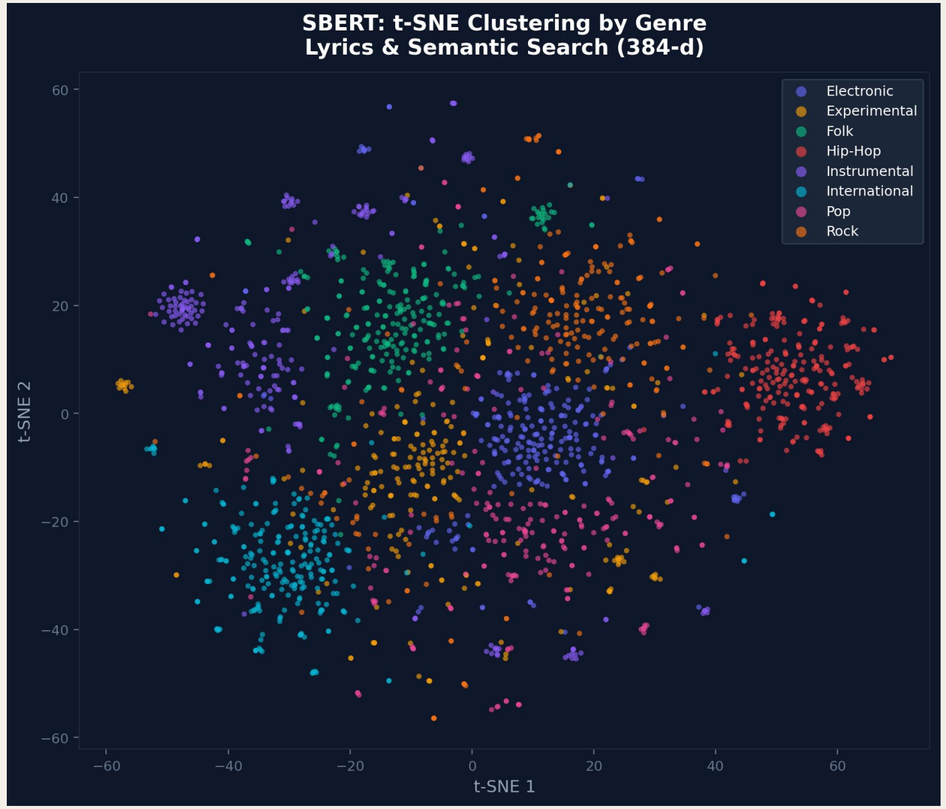
CLAP **LAION CLAP** · HTSAT-tiny backbone, AudioSet+music pretraining
laion/clap

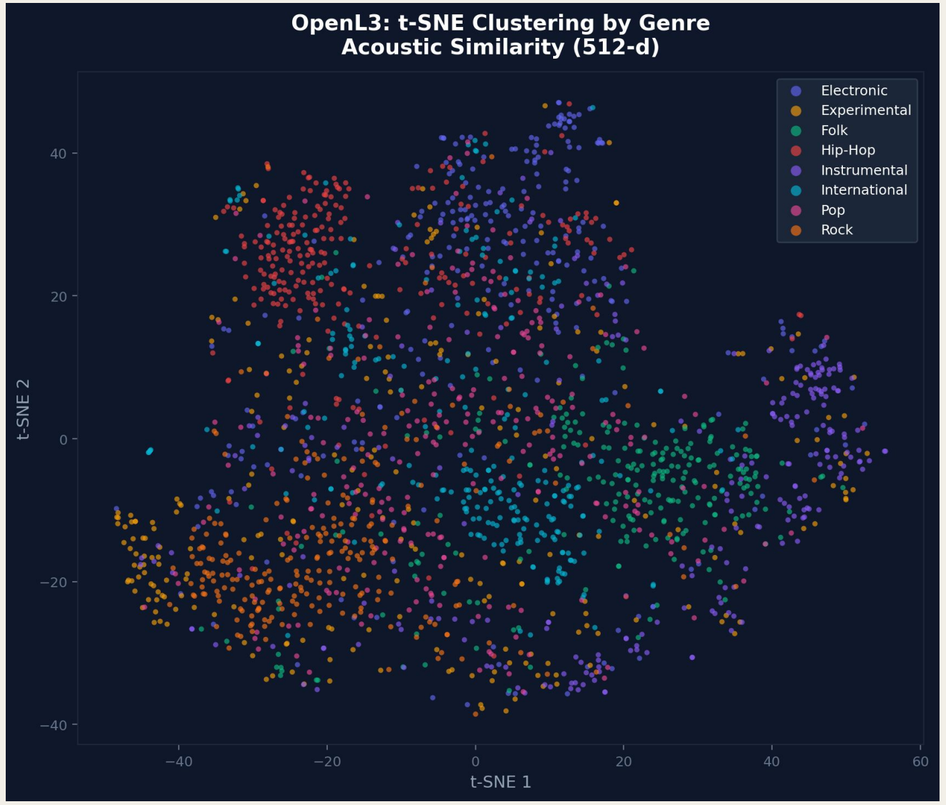
SBERT **Sentence-Transformers** · all-MiniLM-L6-v2 · 384-d
sentence-transformers

L3 **OpenL3** · self-supervised audio embeddings
marl/openl3

PYG **PyTorch Geometric** · heterogeneous GNN (SAGEConv)
pyg-team/pytorch_geometric

FAI **FAISS** · Facebook AI Similarity Search
facebookresearch/faiss


SBERT: t-SNE Clustering by Genre
Lyrics & Semantic Search (384-d)
Electronic
Experimental
Folk
Hip-Hop
Instrumental
International
Pop
Rock
t-SNE 1
t-SNE 2
60
40
20
0
−20
−40
−60


OpenL3: t-SNE Clustering by Genre
Acoustic Similarity (512-d)
Electronic
Experimental
Folk
Hip-Hop
Instrumental
International
Pop
Rock
t-SNE 1
t-SNE 2
40
20
0
−20
−40
60

NOW PLAYING

Oh! Mighty Engine

Neil Halstead · FOLK

0:00 / 0:00

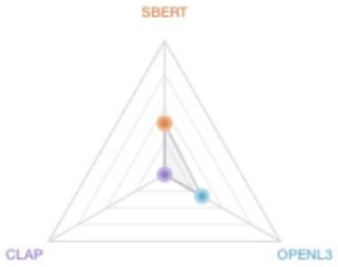
FUSED RECOMMENDATIONS

Reciprocal Rank Fusion

1	One Thing	Brett Saxon · FOLK	0.0400
2	A Penny For Your Dreams	Albin Andersson · FOLK	0.0287
3	In Your Hollow	Allysen Callery · FOLK	0.0282
4	Wayfaring Stranger	David Mumford · FOLK	0.0280
5	Train Coming In	Peter Biedermann · FOLK	0.0269
6	Tonight We Will Rise	Stephen Bartolomei · FOLK	0.0262
7	The Hermit's View	Itasca · FOLK	0.0245
8	Dust Devils	Michael Chapman · FOLK	0.0236
9	Walking In The Country	David Mumford · FOLK	0.0227
10	Merry-Go-Round	Howie & Ann Mitchell · FOLK	0.0164

Merry-Go-Round

Howie & Ann Mitchell



0.383	0.323	-0.095
SBERT	OPENL3	CLAP

SBERT · OPENL3 · CLAP

	SBERT	OPENL3	CLAP
1	Merry-Go-Round Howie & Ann Mitchell		0.383
2	Henry Martin Howie Mitchell		0.374
3	Irish Washerwoman Howie & Ann Mitchell		0.373
4	God speed the Plow The Fucked Up Beat		0.358
5	Living on the Wind Andy G. Cohen		0.332
6	Dona Nobis Pacem Howie Mitchell & Ruth Meyer		0.315

/ THANK YOU

QUESTIONS?

/ DEMO

/ SOURCE

github · multi-faceted-music-retrieval

/ TEAM

Wenny · Sid · Issac · MJ · Jiayi · Helena